

the area, the Tista has unmanageable waters between June and September because of the monsoon and meltwater from the glaciers. It is also susceptible to sudden course changes and flooding, with waves that rise as high as 4.5–6 m. The bridge at Sivok took all this into consideration. Wells, nearly 14-m-deep, were sunk into the riverbed and the structure that was built had one central span of about 76 m and two 45 m spans on either side. When a flood of 0.65 million cusecs breached the east guide bund in June 1950, the bridge was extended by three spans of 45 m on the east.

The Madarihat–Hasimara section was about 14-km-long, but deceptive in its distance, for it involved bridging the Torsa, a fierce river, originating in Tibet and known for its floodwaters. CAH Edwards, engineer-in-chief, surveys and construction, Eastern Bengal Railway, had observed in his 1927 report to the Railway Board that “the velocity of the floods is very great and destructive.” Its width and flotsam from uprooted Sal trees added to its untameable nature. In fact, the earlier line avoided the Torsa with a 320-km detour via Lalmanirhat (a part of East Pakistan, now Bangladesh). In his 1951 book, *A Complete Story of the Assam Rail Link Project*, Karnail Singh wrote that the river would have remained unbridged had it not been for the Partition. Surveyors and engineers had no option but to bridge the Torsa. River training involved using elaborate guide bunds and well foundations sunk to nearly 18 m. The bridge itself was nine spans of 45 m each.

The final section of Alipur Duar to Fakiragram was comparatively uncomplicated, barring a portion of 8 km in Cooch Behar, which involved navigating through the impenetrable, malarial Buxa forests, accessible only by elephants and crossing the Rivers Sankosh and Raidak.

During the course of the construction, 379 channels, spread over 402 km, were bridged, some that remained accessible for only five months in a year. Of these, 22 rivers, including the Tista, Torsa, and Sankosh needed deep well foundations. For the first time, prestressed concrete girders of around 18 m and 12 m spans were used to bridge across the three rivers.